

Problem 22.9

$$\max(\lambda' - \lambda) = \frac{2h}{mc}$$

$$\lambda' - \lambda = \left(\frac{h}{mc} \right) (1 - \cos \phi)$$

$$\max(E_e - mc^2) = \max(E - E')$$

$$\begin{aligned} &= \frac{hc}{\lambda} - \frac{hc}{\lambda + \frac{2h}{mc}} \\ &= E - \frac{E}{1 + \frac{2E}{mc^2}} = \frac{E}{1 + \frac{mc^2}{2E}} \end{aligned}$$

Problem 22.12

$$\Delta E = E_4 - E_3 = -13.6\text{eV}\left(\frac{1}{4^2} - \frac{1}{3^2}\right) = 0.66\text{eV}$$

$$k_{\text{B}} 300\text{K} = \frac{1}{40}\text{eV} = 0.025\text{eV}$$

$$k_{\text{B}} = 8.6 \times 10^{-5}\text{eVK}^{-1}$$

Problem 23.6

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T \sim \frac{3}{2} \frac{1}{40} \text{ eV} \quad \overline{mv^2} = \frac{3}{40} \text{ eV}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{m^2 \overline{v^2}}} = \frac{hc}{\sqrt{mc^2 m \overline{v^2}}} \quad E_k \ll m_0 c^2$$
$$= \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{4 \times 931 \times 10^6 \times 3/40 \text{ eV}}} = 7.4 \times 10^{-2} \text{ nm}$$

The molar volume at STP is

$$V_m = 22.4 \times 10^{-3} \text{ m}^3/\text{mol}.$$

If average spacing is a .

$$\frac{4}{3}\pi a^3 \times N_A = V_m$$

$$a = \left(\frac{3V_m}{\pi N_A} \right)^{1/3} = \left(\frac{3 \times 22.4 \times 10^{-3}}{\pi \times 6.022 \times 10^{23}} \right)^{1/3} = 3.3 \times 10^{-9} \text{ m}$$

So the wavelength is about 2% of the spacing.
Fairly the helium atoms can be treated as particles

Problem 23.8 In a spherical region of diameter 1m, uncertainty in x , Δx , is 1m

$$\Delta p \geq \frac{\hbar}{\Delta x};$$

$$\begin{aligned}\Delta v &= \frac{\Delta p}{m} \geq \frac{\hbar}{m\Delta x} = \frac{\hbar c}{mc^2} \frac{c}{\Delta x} \\ &= \frac{1}{2\pi} \frac{1240\text{eV} \cdot \text{nm}}{938.3\text{MeV}} \frac{3 \times 10^8 \text{ m/s}}{1\text{m}} = 0.063 \times 10^{-6} \text{ m/s}\end{aligned}$$

$$v \geq \Delta v; \quad \min v = 6.3 \times 10^{-8} \text{ m/s}$$

It is quite smaller than daily speed.

Problem 24.3

If $P(x) = c$, the normalization requires

$$1 = \int_0^a c dx, \quad c = \frac{1}{a};$$

$$\langle x \rangle = \int_0^a \frac{1}{a} x dx = \frac{1}{2a} \cdot a^2 = \frac{a}{2}$$

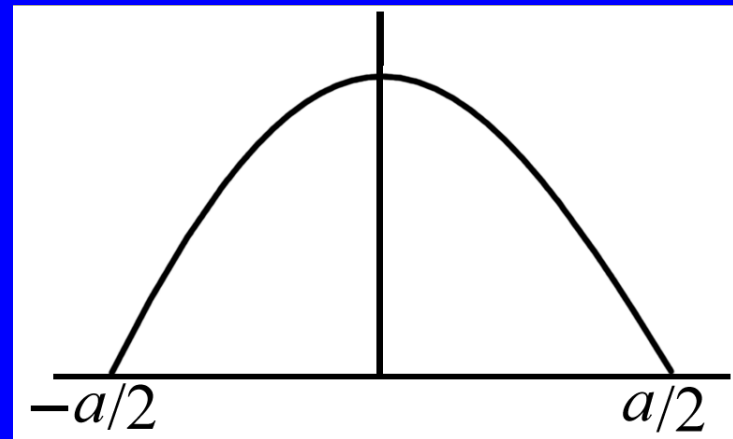
$$\langle x^2 \rangle = \int_0^a \frac{1}{a} x^2 dx = \frac{1}{3} a^2$$

$$\langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2$$

$$\Delta x = \sqrt{\langle (x - \langle x \rangle)^2 \rangle} = \sqrt{\frac{1}{3} a^2 - \left(\frac{a}{2}\right)^2} = \frac{\sqrt{3}}{6} a$$

Problem 24.4

(a)



(b) For symmetric coordinates, the solutions of Schrödinger equation are divided as two groups:

$$n = \text{odd} \quad \psi_n = A \cos\left(\frac{n\pi}{a} x\right);$$

$$n = \text{even} \quad \psi_n = A \sin\left(\frac{n\pi}{a} x\right)$$

So the quantum number is $n = 1$.

(c) Since the wave function is normalized, we have

$$\langle x \rangle = \int_{-a/2}^{a/2} \psi(x) x \psi(x) dx = 0$$

$$\langle x^2 \rangle = \int_{-a/2}^{a/2} \psi(x) x^2 \psi(x) dx = \left(\frac{1}{12} - \frac{1}{2\pi^2} \right) a^2$$

$$\Delta x = a \sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}$$

If the wave function is not normalized, we have

$$\langle Q \rangle = \frac{\int \psi^*(\mathbf{r}) \hat{Q} \psi(\mathbf{r}) d\tau}{\int \psi^*(\mathbf{r}) \psi(\mathbf{r}) d\tau}$$

Problem 24.5

For potential step, we have

$$\psi = Ae^{ikx} + Be^{-ikx} \quad (x < 0)$$

$$\psi = Ce^{ik_1x} \quad (x > 0)$$

Matching at $x = 0$,

$$A + B = C$$

or

$$k(A - B) = k_1C$$

$$\frac{C}{A} - \frac{B}{A} = 1$$

$$\frac{k_1}{k} \frac{C}{A} + \frac{B}{A} = 1$$

$$\frac{C}{A} = \frac{2k}{k + k_1}; \quad \frac{B}{A} = \frac{k - k_1}{k + k_1}$$

$$T = \frac{4k^2}{(k + k_1)^2} \frac{k_1}{k}; \quad R = \left(\frac{k - k_1}{k + k_1} \right)^2$$

Eq.(24.4.16)

Without a factor of k_1/k in T , $R + T = 1$ will be violated

Problem 24.6

$$\psi_0 = \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} \quad \psi_1 = \left(\frac{\alpha}{2\sqrt{\pi}} \right)^{1/2} 2\alpha x e^{-\frac{1}{2}\alpha^2 x^2};$$

$$\int_{-\infty}^{\infty} \psi_0^2 dx = \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\alpha^2 x^2} dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt = 1$$

Appendix G :
Gaussian integral

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} \psi_1^2 dx &= \frac{\alpha}{2\sqrt{\pi}} \int_{-\infty}^{\infty} 4\alpha^2 x^2 e^{-\alpha^2 x^2} dx \\
 &= \frac{2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = 1
 \end{aligned}$$

Appendix G :
Gaussian integral

$$I_n = \int_0^{\infty} x^n e^{-x^2} dx = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right)$$

$$\int_{-\infty}^{\infty} \psi_0 \psi_1 dx = \frac{\alpha}{\sqrt{2}\sqrt{\pi}} \int_{-\infty}^{\infty} 2\alpha x e^{-\alpha^2 x^2} dx = 0$$

odd function

$$\psi_2 = \left(\frac{\alpha}{2^3 \sqrt{\pi}} \right)^{1/2} (4\alpha^2 x^2 - 2) e^{-\frac{1}{2}\alpha^2 x^2};$$

$$\int_{-\infty}^{\infty} \psi_0 \psi_2 dx = \frac{\alpha}{2\sqrt{2}\sqrt{\pi}} \int_{-\infty}^{\infty} (4\alpha^2 x^2 - 2) e^{-\alpha^2 x^2} dx$$

$$= \frac{1}{2\sqrt{2}\sqrt{\pi}} \int_{-\infty}^{\infty} (4t^2 - 2) e^{-t^2} dt$$

$$= \frac{1}{2\sqrt{2}\sqrt{\pi}} \left[4\Gamma\left(\frac{1}{2}\right) - 2\sqrt{\pi} \right] = 0$$

Problem 24.7

(a) $\nu = 400\text{Hz}$

$$\begin{aligned} E_0 &= \frac{1}{2} h \nu \\ &= \frac{1}{2} \times 6.626 \times 10^{-34} \text{ J/s} \times 400\text{Hz} \\ &= 1.33 \times 10^{-31} \text{ J} \end{aligned}$$

$$\Delta E = h \nu$$

(b)

$$\nu = 400\text{Hz}$$

$$\begin{aligned} E_0 &= \frac{3}{2} h \nu \\ &= \frac{3}{2} \times 6.626 \times 10^{-34} \text{ J/s} \times 400\text{Hz} \\ &= 3.98 \times 10^{-31} \text{ J} \end{aligned}$$

$$\Delta E = h \nu$$

(c)

$$\nu = 6.43 \times 10^{11} \text{ Hz};$$

$$\begin{aligned} E_0 &= \frac{1}{2} h \nu \\ &= \frac{1}{2} \times 6.626 \times 10^{-34} \text{ J/s} \times 6.43 \times 10^{11} \text{ Hz} \\ &= 2.13 \times 10^{-22} \text{ J}; \end{aligned}$$

$$\Delta E = h \nu$$